# 1 Quasiparticle-interference results for a two-dimensional superconducting multiorbital Weyl SSH model

The emergence of topological superconductivity in a two-dimensional (2D) Weyl system, composed of stacked Su-Schrieffer-Heeger (SSH) chains was studied by Rosenberg and Manousakis [CITE: Rosenberg, Aryal PRB 100 104522 (2019)]. They showed that an attractive Hubbard interaction between spinful electrons leads to a topological trivial supercondictor. Next, a pairing interaction that couples spinless fermions on opposite sublattices within the same unit cell was shown [CITE: Rosenberg and Manousakis arXiv:2203.12004v1] to induce a topological superconducting state, characterised by a gap function with a non-trivial phase, as well as Majorana and Fermi arc edge states. We reproduce the results in a finite model with open-boundary conditions along the direction of SSH dimerisation and periodic boundary condition on the perpendicular axis. We then investigate the model with an impurity, self-consistently, in the mean-field limit. We show the expected quasiparticle-interference, superconducting gap spectrum and local density of states in such a system. Later, we also wish to investigate $v = w = t_d$, for the purposes of spontaneous Pierls or Jahn-Teller distortion (what is this? CDW?), as well as a system with periodic-boundary conditions and a wall of impurities.

## 1.1 Reproducing results

The Hamiltonian is given by

$$\begin{aligned}\hat{\mathcal{H}}_0 &= -\sum_{1\leq x,y\leq N}\left[v\hat{c}^{(A)\dagger}_{x,y}\hat{c}^{(B)}_{x,y} + w\hat{c}^{(A)\dagger}_{x,y}\hat{c}^{(B)}_{x-1,y} + \text{h.c.}\right] \\ &\quad -\sum_{1\leq x,y\leq N}\left[t_d\hat{c}^{(B)\dagger}_{x,y}\hat{c}^{(A)}_{x,y\pm1} + t_d\hat{c}^{(B)\dagger}_{x,y}\hat{c}^{(A)}_{x+1,y\pm1} + \text{h.c.}\right] \\ &\quad -\sum_{1\leq x,y\leq N}\left[\left(\mu + s(-1)^m\right)\hat{n}^{(m)\dagger}_{x,y}\right], \end{aligned} \tag{1}$$

where the fermion operator $\hat{c}^{(m)\dagger}_{x,y}$ creates an operator on atom $m \in \{A, B\}$ within the unit cell located at $\mathbf{R}_{x,y} = x\mathbf{b}_0\hat{x} + y\mathbf{b}_1\hat{y}$, where $\mathbf{b}_0, \mathbf{b}_1$ are the lattice basis vectors, $\hat{n}^{(\alpha)\dagger}_{x,y}$ is the number operator, $\mu$ is the chemical potential and $s$ is a rigid shift of the atomic potentials. The intra- and inter-unit-cell hoppings strengths in the $\hat{x}$ direction are given by $v$ and $w$, which form the terms of the standard SSH model. These 1D SSH chains are coupled by a diagonal hopping $t_d$, which form a two-dimensional stacking. The lattice is depicted with $N = 3$ unit cells in Figure 1. Following [CITE: Rosenberg and Manousakis arXiv:2203.12004v1], without loss of generality, we set $v = 0.6, w = 1.2$, the topological phase of the SSH model, and $t_d = 0.9$.

The solution to the Hamiltonian is given by the Green's function, which we remind the

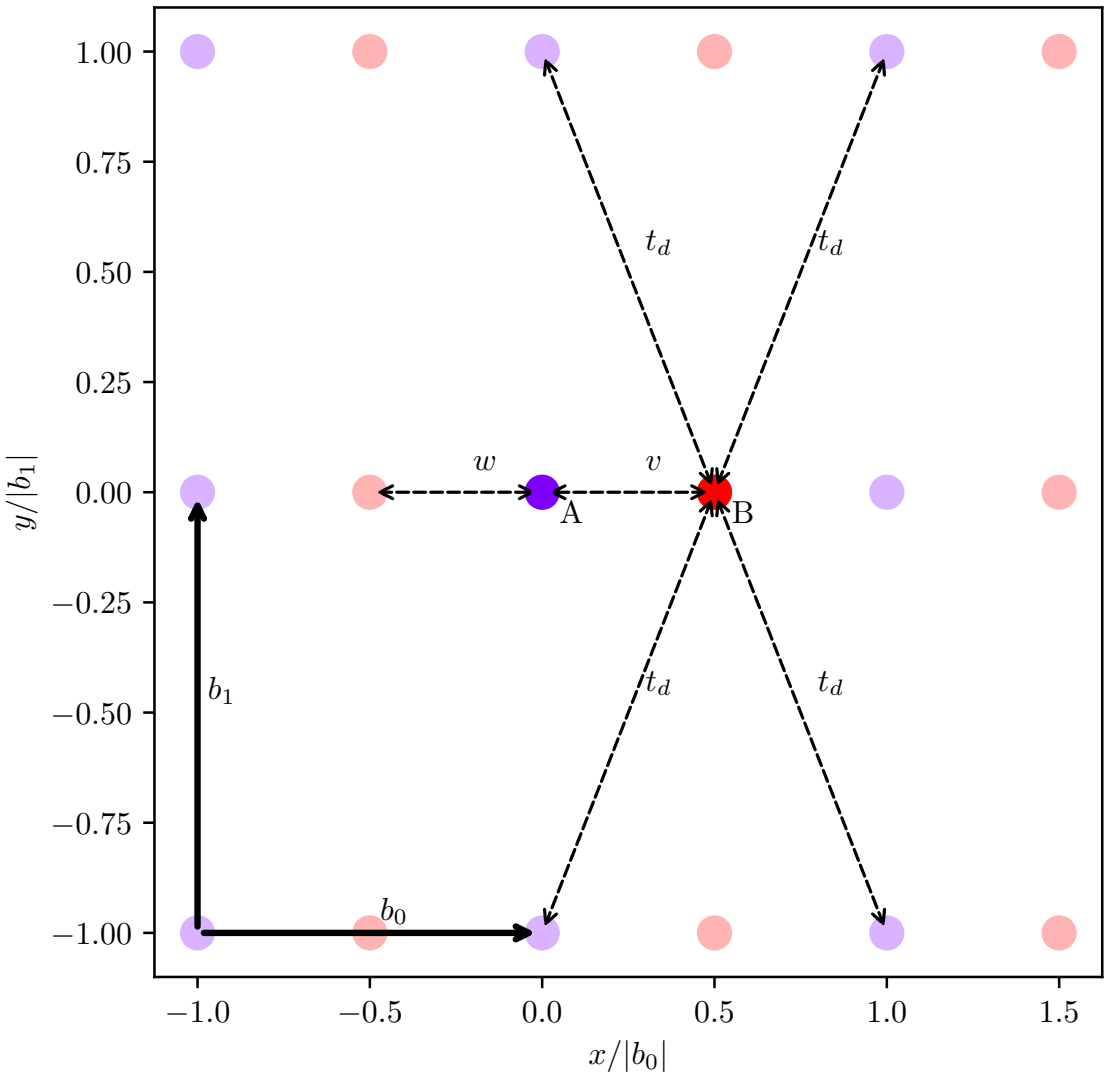


Figure 1: Two-dimensional multiorbital Weyl SSH model. Along the $\hat{x}$-direction, atomic sites A and B within the unit cell form an SSH chain, with intracell hopping $v$ and intercell hopping $w$. An additional interchain hopping parameter couples the chains into a two-dimensional stack. Following Rosenberg and Manousakis, we study a diagonal hopping $t_d$, which is a Type II extension of the SSH model in the classification according to Bo-Hung [CITE: Two-Dimensional Extended Su-Schrieffer-Heeger Model; Masters Thesis; Chen, Bo-Hung; July 2018]. The basis for the lattice is given by the vectors $b_0$ and $b_1$.

reader, is given by

$$\hat{\mathcal{G}}(\omega - i\epsilon, x, y) \equiv \sum_n \left[ \frac{\psi_n(x,y)\psi_n(x,y)^\dagger}{\omega - i\epsilon - E_n} \right], \tag{2}$$

where $\psi_n(x, y)$ is the eigenvector associated with the $n^{\text{th}}$ eigenenergy, resolved at the point on the lattice $(x, y)$. The Green's function matrix has a component for each atom. The Green's function is experimental observables through the *local density of states*,

$$\text{LDOS}(\omega; x, y) \equiv -\frac{1}{\pi} \Im \hat{\mathcal{G}}(\omega - i\epsilon, x, y), \tag{3}$$

which is a measure of the density of states at a given point, at a specified applied bias energy $\omega$ and resolution $\epsilon$. The local density of states for each atomic site with the unit cells are given in Figure 2, showing the existence of zero energy edge states.

Resolving the local density states at the level of the unit cell (summing over the atoms) and integrating over the $y$-axis, we study the *spectrum* of the system. The spectrum shows edge modes present at zero energy at the edges, which are not present in the bulk. These are distinct signatures of Majorana modes. In addition to localised density at the edge, there exists a 'tail' of density which the authors investigate further.

Majorana modes exist pinned to *Dirac points*, which are the cross point of two bands. Studying the band structure in Figure 10, we observe a zero-mode between the Dirac points, which is known as a *Bogoliubov-Fermi arc*.

In order to determine whether the modes are topological modes, it must be shown that they their wavefunctions are distributed nonlocally across the system. We plot the wavefunction via the related *spectral density* of the Green's function at the point $k_y$ at which the Bogoliubov-Fermi arc and Majorana modes are are located, Figure 8. As a function of $x$-position, we observe that indeed, the modes are spread non-locally and that the arc tails.

Mention Bo Huang's thesis results on the influence of the shape of the lattice on the topological modes.

### 1.2 Interactions and mean-field solution

We add an interaction term to model multiorbital superconductivity within the unit-cell, $U_v$, and an interorbital intercell Coulomb repulsion term, $U_w$. The interacting contribution to the Hamiltonian is

$$\hat{\mathcal{H}}_I = -U_v \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)\dagger}_{x,y} \hat{c}^{(A)}_{x,y} \hat{c}^{(B)}_{x,y} + U_w \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)\dagger}_{x-1,y} \hat{c}^{(B)}_{x-1,y} \hat{c}^{(A)}_{x,y}. \tag{4}$$

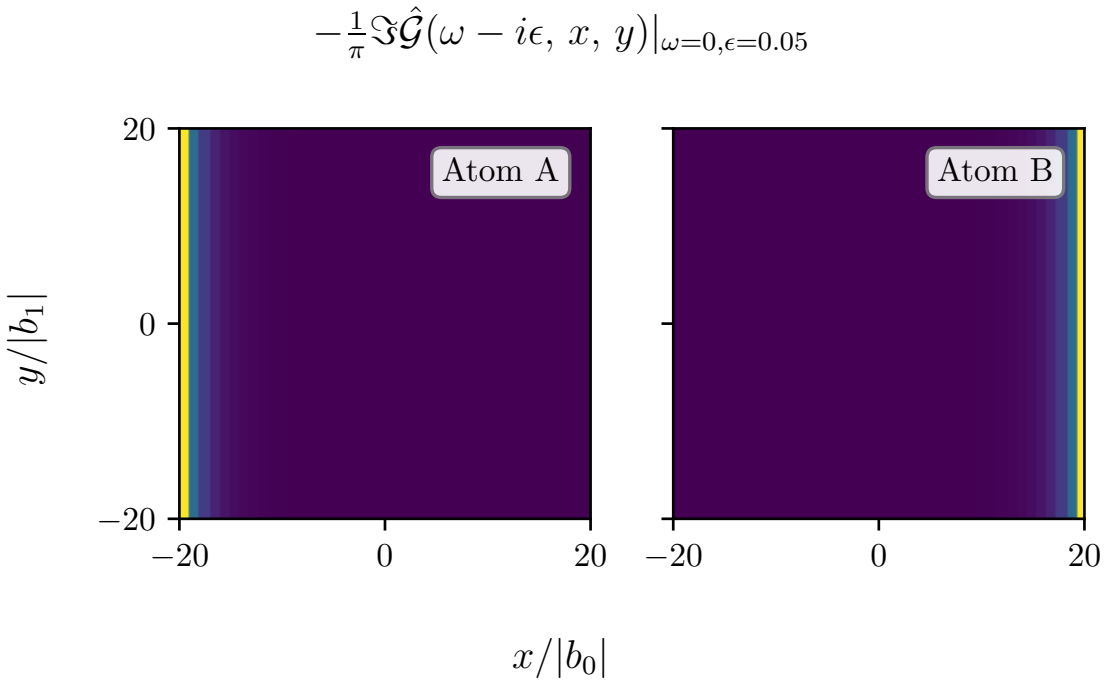


Figure 2: The local density of states of a Weyl SSH model, that is, a spinless square lattice with $41 \times 41$ lattice cells, with chemical potential $\mu/t = 0.00$, zero shift $s = 0$ and applied bias $\omega = 0$ and spectral resolution $\epsilon = 0.05$. The SSH chains are in their topological phase $w = 1.2 > v = 0.6$, with diagonal hopping $t_d = 0.9$, which does gives rise to no topological modes perpendicular to the chains. Notice the zero bias density at the left edge for atom A and on the right for atom B, a signature of topological modes.

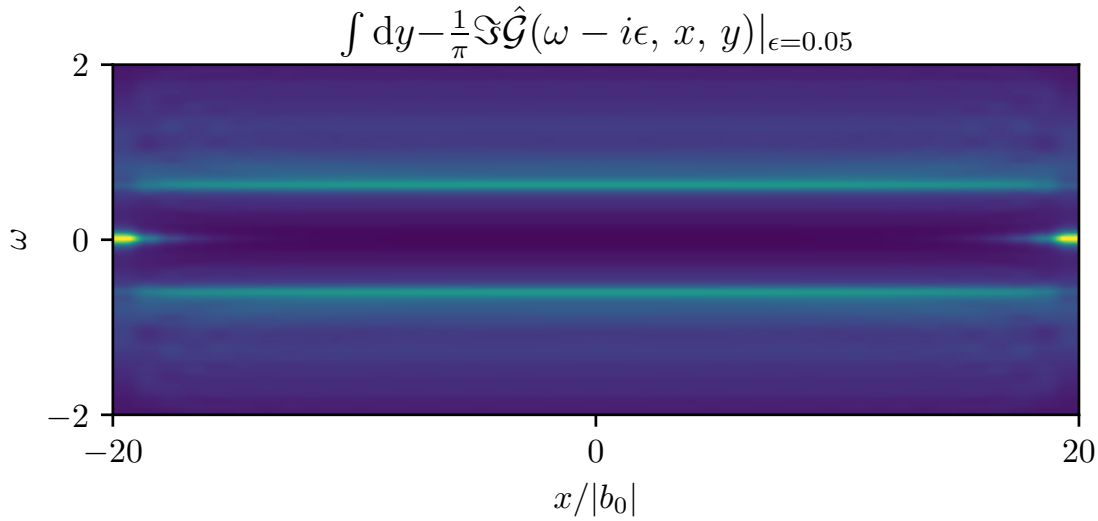


Figure 3: The spectrum of the Weyl-SSH model integrated over the direction perpendicular to the SSH chains and summed over the atomic sites within the lattice cell. Zero-energy modes are observed at the edges, with a faint tail of density.

The mean-field Hamiltonian is

$$\hat{\mathcal{H}}_I' = \sum_{x,y} \Bigg[ \sum_m \left[ \phi_m \hat{n}_{x,y}^{(m)} \right] + \Phi_v \hat{c}_{x,y}^{(A)} \hat{c}_{x,y}^{(B)} + \Phi_w \hat{c}_{x,y}^{(A)} \hat{c}_{x-1,y}^{(B)} + \Delta_v \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x,y}^{(B)\dagger} + \Delta_w \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x-1,y}^{(B)\dagger} + \Delta_v^* \hat{c}_{x,y}^{(A)} \hat{c}_{x,y}^{(B)} + \Delta_w^* \hat{c}_{x,y}^{(A)} \hat{c}_{x-1,y}^{(B)} \Bigg], \quad (5)$$

where $\phi_v$ is the *Hartree* term which renormalises the chemical potential, allowing for a Peirl's instability. $\Phi_v$ is a *Fock* term, describing renormalistion of the hopping parameter $v$. The additional *Gorkov* terms $\Delta_v$ capture fermion-fermion pairing, relevant to theories of magnetism and superconductivity. These terms are known as the method of *(superconducting) density functional theory, SC-DFT*. The self-consistent solution to the mean field

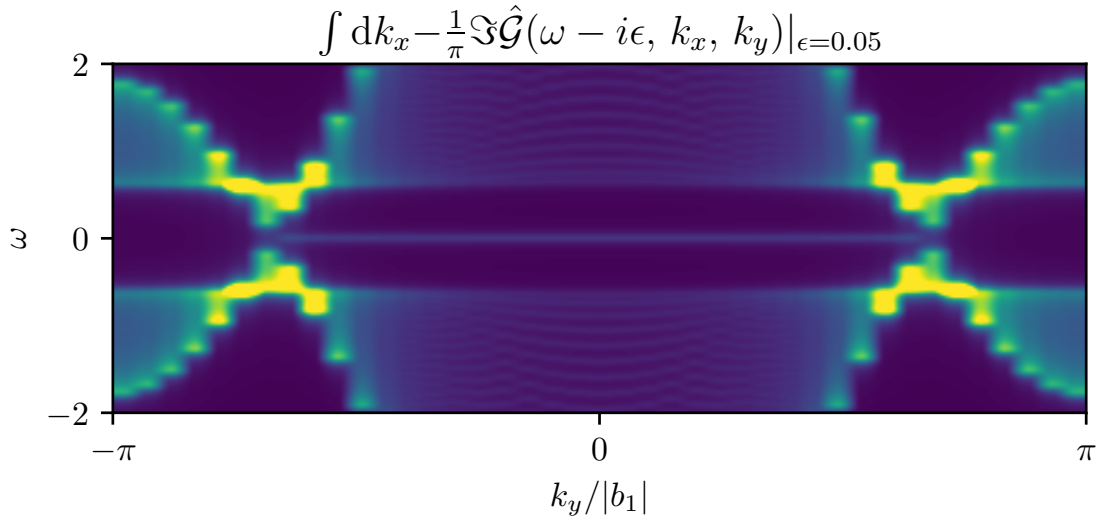


Figure 4: The spectrum of the Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. Dirac nodes are observed, which pin Majorana zero-modes. Between the Dirac nodes is a line of zero-energy states, known as a Fermi-Bogoliubov arc.

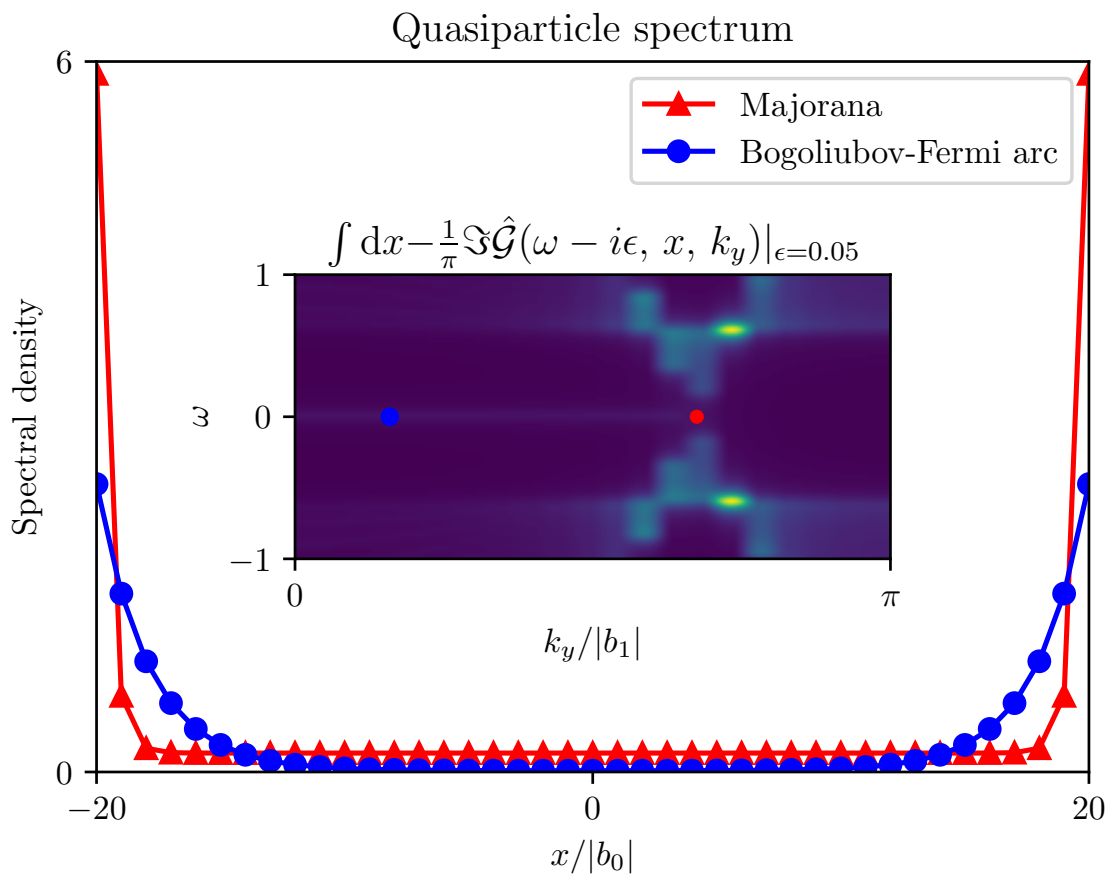


Figure 5: The spectrum of the Majorana and Bogoliubov-Fermi arc quasiparticle modes, at zero-applied bias, with the vertical momentum specified by the (colour-coordinated) spectral plot. The quasiparticle density is plotted over the direction parallel to the SSH chains. The Majorana mode (red) is localised at the edges. The Bogoliubov-Fermi arc, in contrast, tails.

is the saddle point $\langle \hat{\mathcal{H}}_I \rangle = \langle \hat{\mathcal{H}}_I' \rangle$. The solutions are

$$\rho_m = (-U_v + U_w))\langle \hat{n}_{x,y}^{(m')} \rangle, \tag{6}$$

$$\Phi_v = -U_v \langle \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x,y}^{(B)} \rangle, \tag{7}$$

$$\Phi_w = U_w \langle \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x-1,y}^{(B)} \rangle, \tag{8}$$

$$\Delta_v = +U_v \langle \hat{c}_{x,y}^{(A)} \hat{c}_{x,y}^{(B)} \rangle, \tag{9}$$

$$\Delta_w = -U_w \langle \hat{c}_{x,y}^{(A)} \hat{c}_{x-1,y}^{(B)} \rangle, \tag{10}$$

$$\Delta_v^* = +U_v \langle \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x,y}^{(B)\dagger} \rangle, \tag{11}$$

$$\Delta_w^* = -U_w \langle \hat{c}_{x,y}^{(A)\dagger} \hat{c}_{x-1,y}^{(B)\dagger} \rangle, \tag{12}$$

where $m'$ refers to the opposite atom to $m$. Note the sign of the Gorkov terms due Fermionic anticommutation.

For simplicity and because we are in particular interested in the multiorbital pairing due to $U_v$, we neglect the intercell interaction $U_w$.

### 1.3 Results with impurities

The impurity is located at the centre with coupling strength $V = 1.21$.

The $x$-axis is glued to periodic boundary conditions.

Our investigation shows no interesting interplay between impurities and topological modes. However, impurities can be used to generate quasiparticle-interference patterns as an experimental diagnosis of topological modes.

### 1.4 Quasiparticle-interference patterns for Majorana and Bogoliubov-Fermi arcs

### 1.5 Future directions

- Phase diagram of self-consistent calculation of bulk Weyl-SSH model.
- Think of a way to pair two (spinless) Weyl-SSH models, via some spin-coupling, such that a regime in which one spin-chain is superconducting and the other is normal can be engineered.
- Publish code.
- Present QPI results at *ExoSup*.

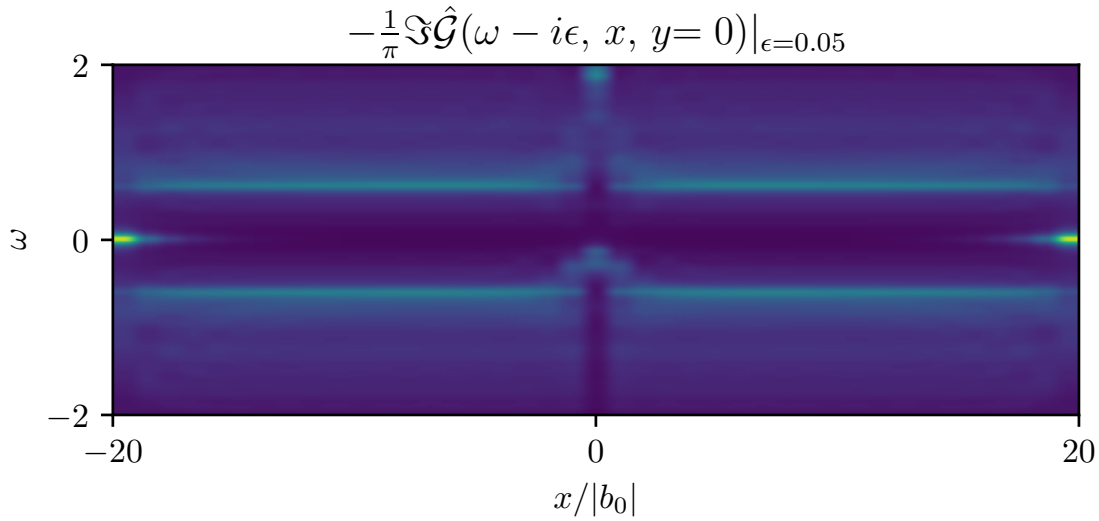


Figure 6: The local density of states along a line parallel to the axis of the dimers through a central impurity. The chemical potential is $\mu/t = 0.00$ and the lattice is $41 \times 41$ with periodic boundary conditions along the axis perpendicular to the dimers, and open boundary conditions along the axis. The impurity has coupling strength $V/t = 1.21$. No additional zero-mode is seen.

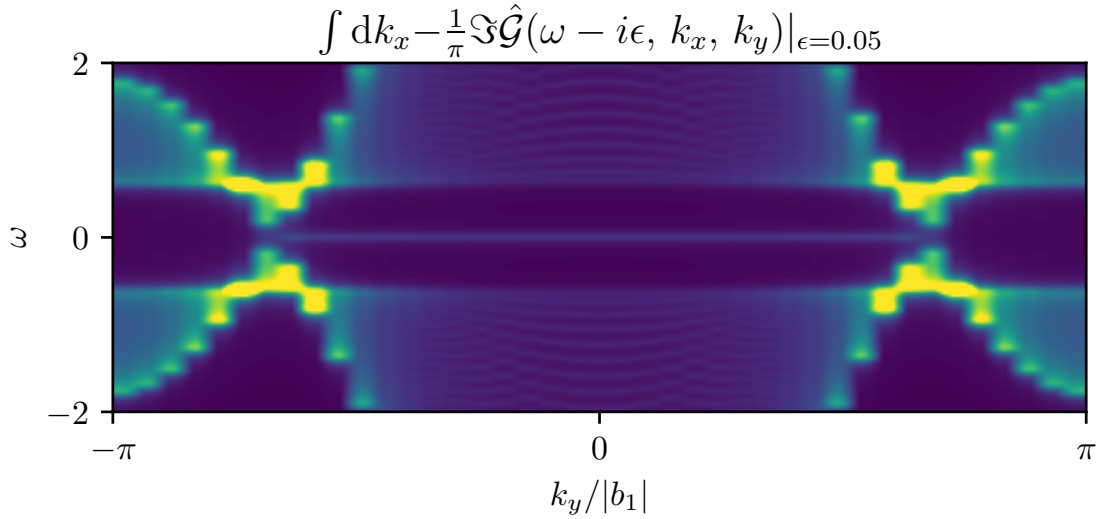


Figure 7: Band structure integrated over the axis of the SSH dimers, $k_x$, with open boundary conditions along the axis perpendicular to the dimers. No qualitative change in the spectrum is observed from the impurity-free model.

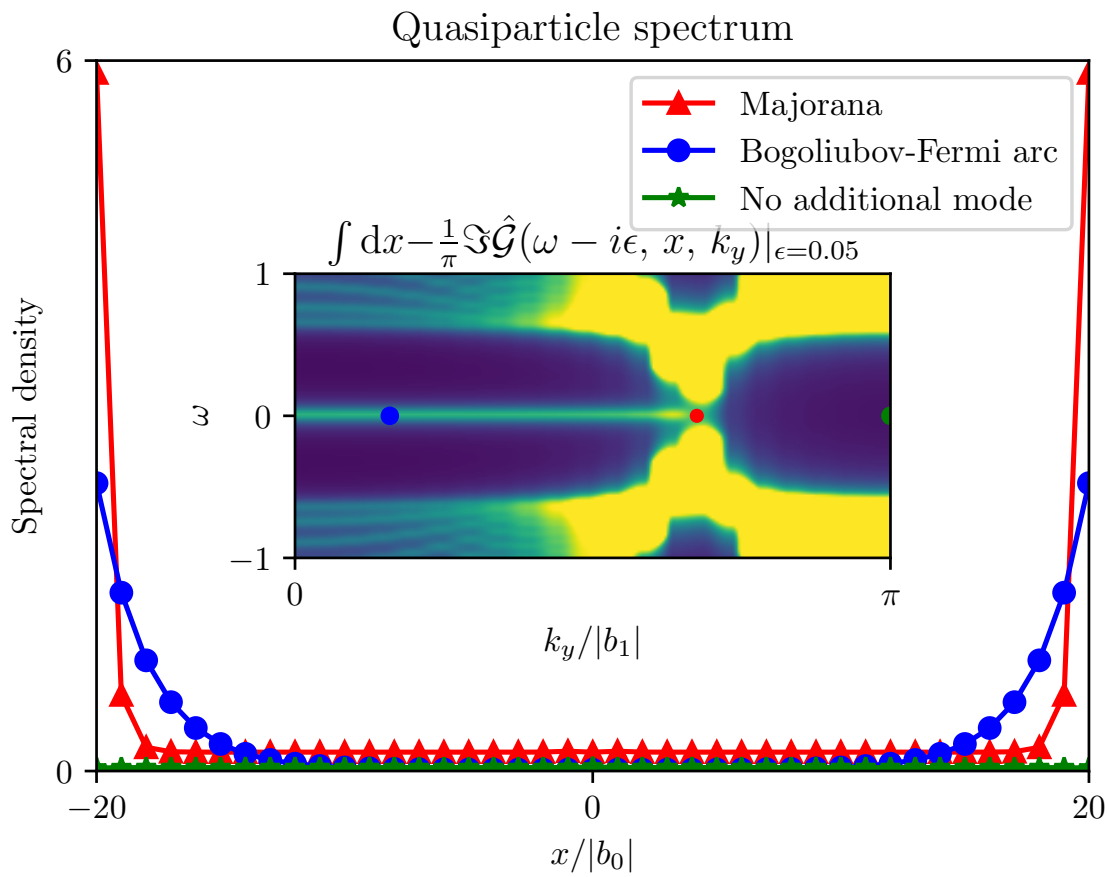


Figure 8: Topological modes in the presence of a single impurity. No localisation of the quasiparticles around the impurity are seen. Compare with Figure 11 –we observe an additional Bogoliubov-Fermi arc in a model with a wall of impurities. Note that the axis perpendicular to the dimers has been integrated over in the spectral density plot.

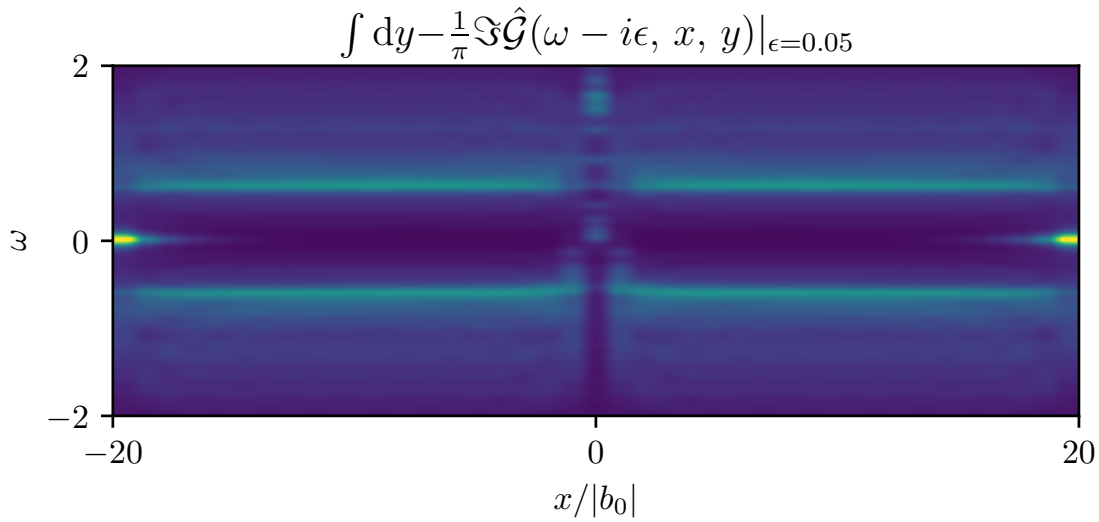


Figure 9: Local density of states with an impurity at the centre. The chemical potential is $\mu/t = 0.00$ and the lattice is $41 \times 41$ with periodic boundary conditions perpendicular to the axis of the dimers, and open boundary conditions along the axis parallel. The impurity wall runs perpendicular to the axis and through the middle, with coupling strength $V/t = 1.21$. Additional modes are seen although at such weak coupling, it is not clear whether they are gapped.

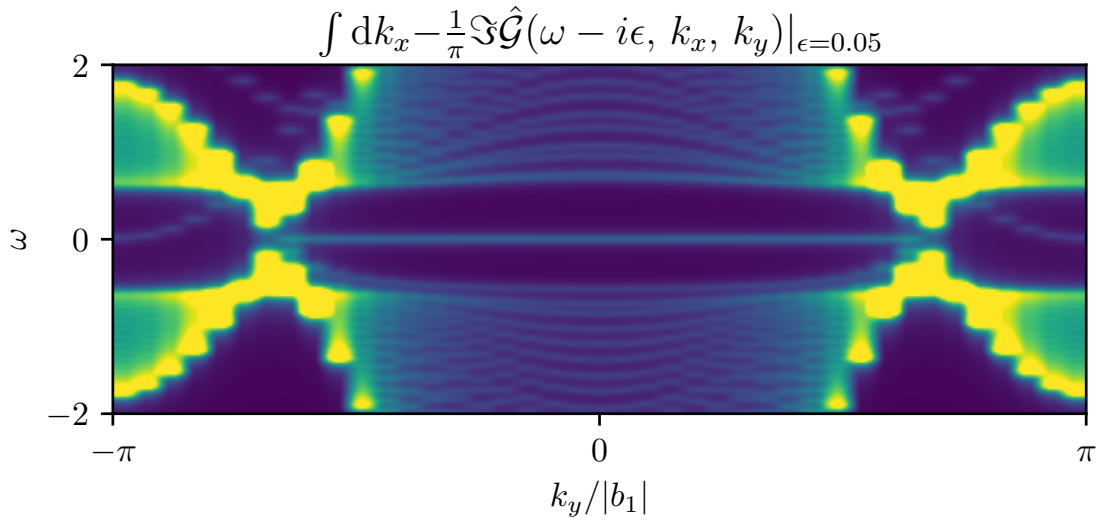


Figure 10: Local density of states integrated over the axis of the SSH dimers and summed over the intracell atomic sites show clearly that there exists an additional gapped mode due to the impurity wall. The chemical potential is $\mu/t = 0.00$ and the lattice is $41 \times 41$ with periodic boundary conditions along the axis perpendicular to the dimers, and open boundary conditions parallel. The impurity wall has coupling strength $V/t = 1.21$.

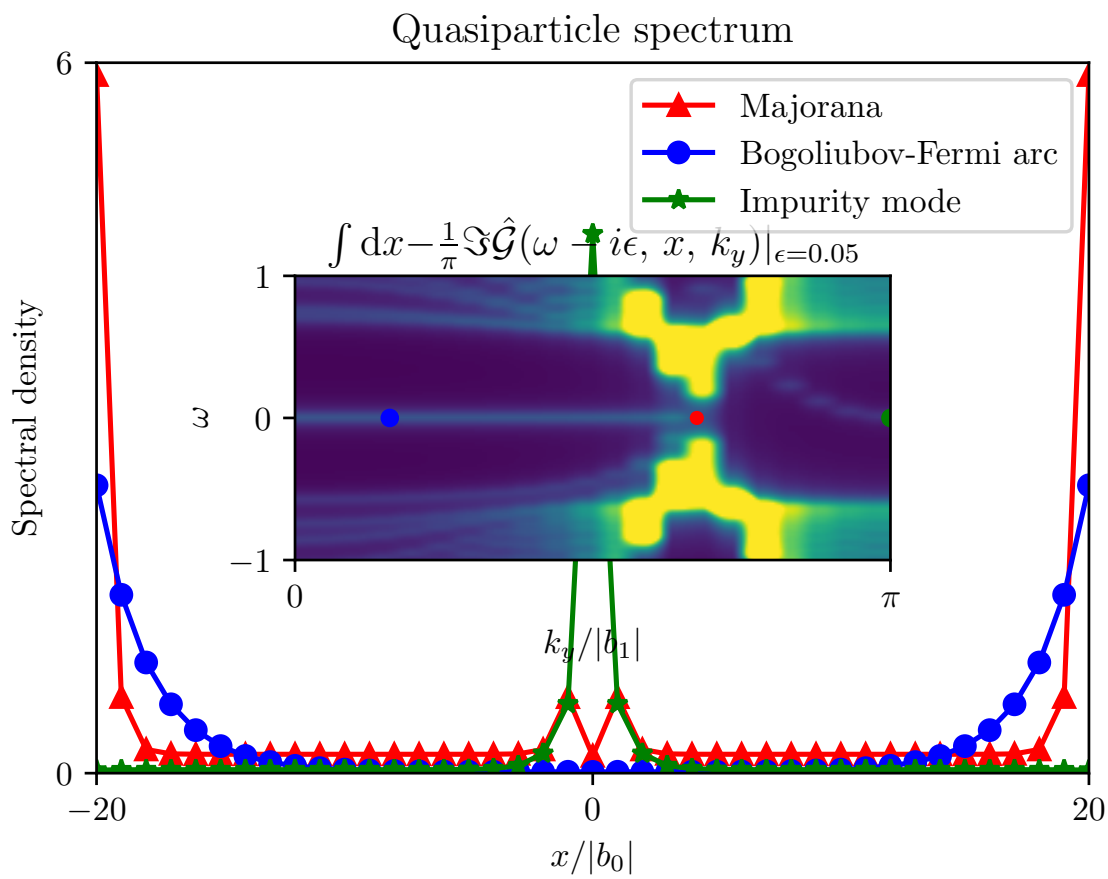


Figure 11: The majorana quasiparticle exhibits some pinning around the impurity wall and an additional arc mode exists at $k_y = \pi$, whose nature is to be investigated.

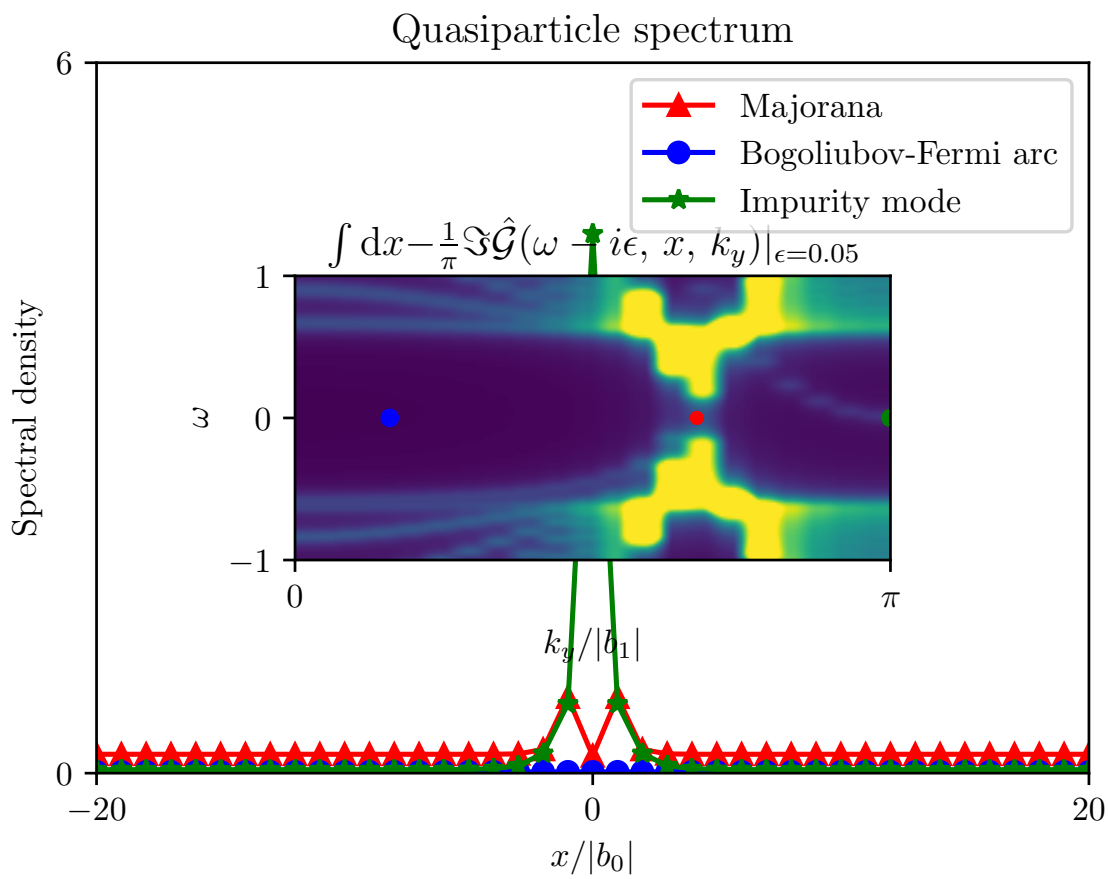


Figure 12: Investigation of the modes with periodic boundary conditions along both axes. The additional mode is shown to be due to the presence of a wall of impurities.

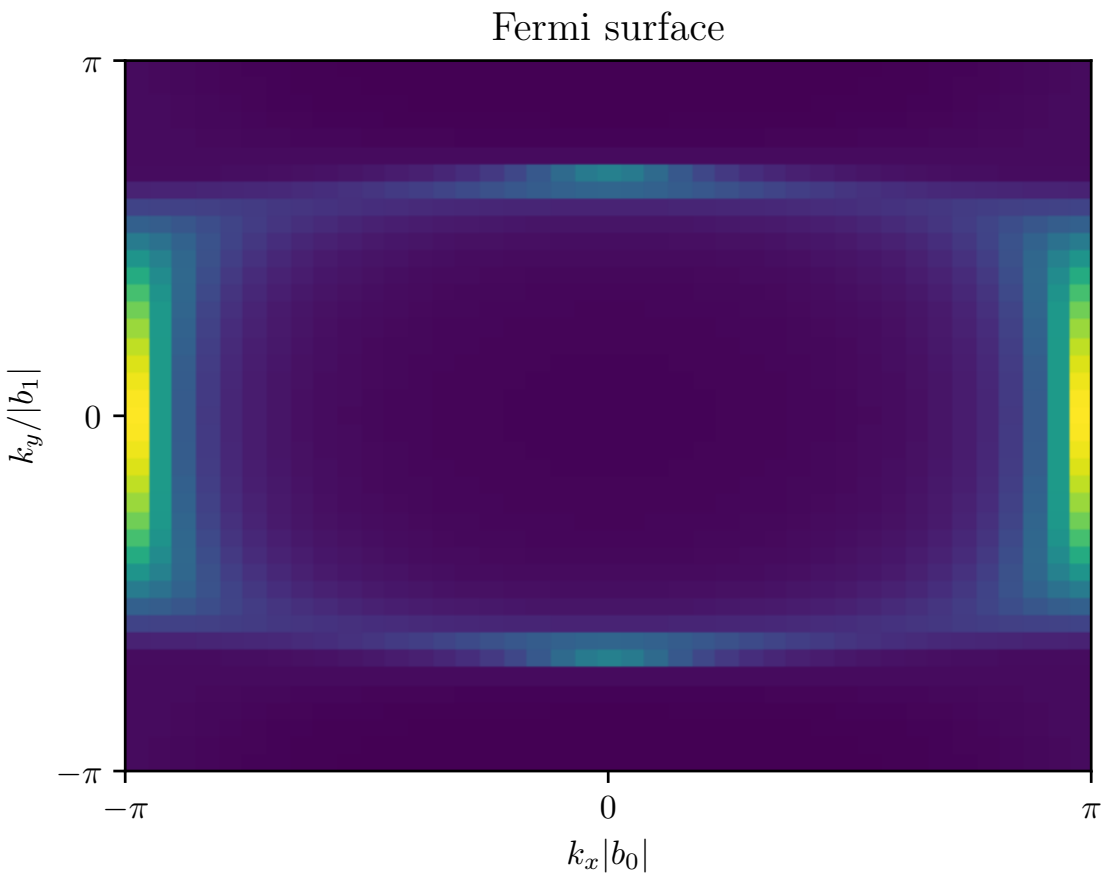


Figure 13: Fermi surface with a central impurity with coupling strength $V = 1.21$ and chemical potential $\mu = 0.0$, in the normal state. Notice density concentrated near $k_y \approx \pi/2$ corresponding to the Majorana mode, and density around $k_x \approx \pm\pi$ corresponding to the Fermi arc.

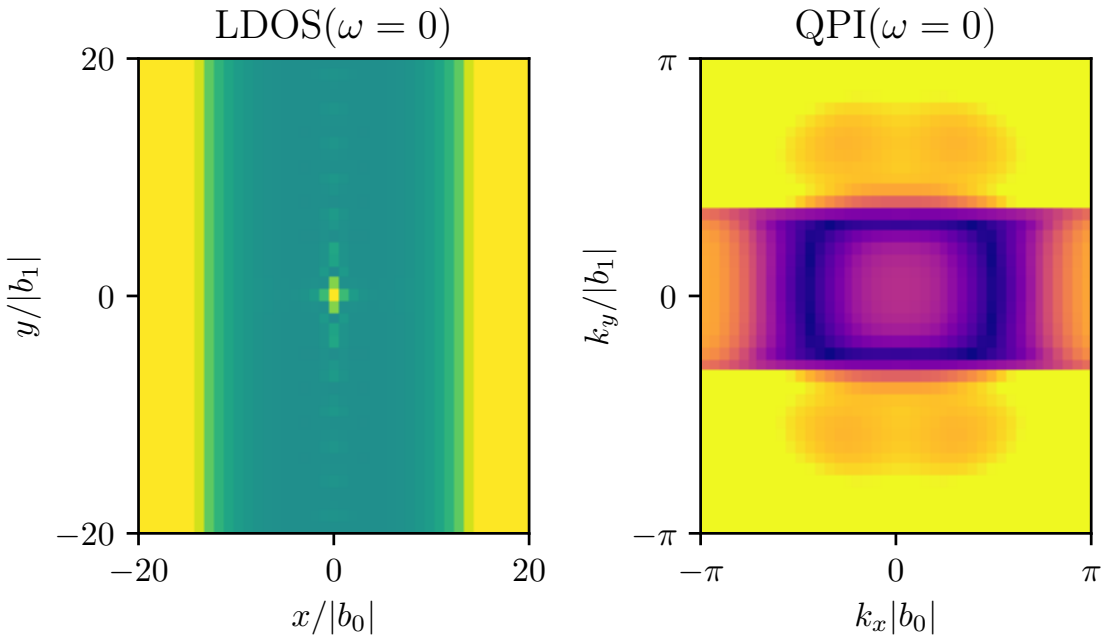


Figure 14: The local density of states and quasiparticle interference pattern for an $41 \times 41$ cell Weyl-SSH model with periodic boundary conditions along the vertical, and open boundary conditions along the horizontal, with a central impurity with coupling strength $V/t = 1.21$. The chemical potential is $\mu/t = 0.00$.